456/1
MATHEMATICS
Paper 1
2026
$2\frac{1}{4}$ hours

Uganda Certificate of Education

MATHEMATICS

Paper 1

2 hours 15 minutes

INSTRUCTIONS TO CANDIDATES:

*This paper consists of **two** sections; **A** and **B.** It has six examination items.*

*Section **A** has **two** compulsory items.*

*Section **B** has **two** parts; **I** and **II.** Answer **one** item from each part.*

*Answer **four** examination items in all.*

*Any additional item(s) answered will **not** be scored.*

***All** answers **must** be written in the answer sheets provided.*

Graph paper is provided.

Silent, non-programmable scientific calculators and mathematical tables with a list of formula may be used.

Turn Over

SECTION A

Answer **both** *items from this section*

Item 1

Three brothers decided to start a school, Kato contributed one hundred and five million shillings ,Kakuru contributed 225million shillings while Kiiza contributed a third of what Kakuru had contributed ,they were advised to spend 60% of their total contributions on building the school structures ,0.3 of the money on school furniture and fittings and save the rest for salaries and miscellaneous expenses for the first term ,so as their business manager they requested you to deposit the money on the account using the ATM machine that was near but the PIN had been saved as 1102 in base four but has to be converted to base five to obtain the exact PIN.

They also agreed to share their profits in the ratio of their contributions at the end of each year. In the first year of operation, they made a profit of 8,100,000 shillings but Kiiza was not sure of what he was supposed to get so he asked for your help.

Task.

(a) Show the brothers how much money you are to deposit on the account and how you generated the actual PIN.

(b) How much of the profits was supposed to go to Kiiza?

Item 2

Susan has opened up a drug shop and needs to buy some drugs P and Q to stock her shop, at one wholesale pharmacy in town she was told that a tin of P and 3boxes of Q would cost shs 45,000 while one tin of P and a box of Q would cost shs 20,000, she needs to determine the cost of each tin of P and each box of Q so that she can set her price at the drug shop.

She also wants to sell her plot that she bought in the neighboring village in order to buy one in the village where her drug shop is however she is not sure of the exact dimensions but she knows that it is rectangular in nature with its length being $\mathbf{20m}$ more than the width with an area $\mathbf{6300m^2}$. She has asked for help to determine the exact dimensions of the plot so as to set its price.

Task

(a) Determine the cost of one tin of P and one box of Q.

(b) Help Susan to determine the dimensions of her plot

SECTION B

Part I

Answer only **one item** *from this part.*

Item 3

At a given hardware shop, the manager requested the owner to increase on the number of casual workers who help to load cement for clients since the customers had now increased. The owner requested the manager to present the record of the sales so as to determine the mean sales per day. The record of the number of bags of cement sold that day to 40 clients was as below.

49	40	45	36	67	32	45	42
40	42	61	48	45	67	51	53
35	46	50	43	47	37	47	47
49	30	62	48	53	41	53	43
45	49	51	44	48	38	49	48

The owner needs this information presented a graphically to make a decision. If 75% of the clients bought more than 51 bags of cement, a new worker will be brought to the shop.

Task

(a) Represent the information on a frequency distribution table with equal class boundaries.

(b) (i) Represent the above information on a statistical graph which can be used to determine if a new worker is to be brought to the hardware shop.

(ii) Will the new worker be brought to the hardware shop? Give a reason

Item 4

In order to improve on food security in a certain village, 130 farmers decided to grow crops which included Maize, Beans, Cassava and others. 62 famers decided to grow Beans, 83 farmers to grow Maize and 60 farmers for Cassava. 30 farmers were to grow both beans and Maize, 40 were to grow both Maize and Cassava while 20 farmers were to grow both beans and cassava. 18 farmers decided to grow only Maize and some famers were to grow other crops other than food crops. It is known by the farmers that if 50% of them do not grow more than one of the above food crops, some members in the village will no get food. A certain organization wishes to reward **one** of the farmers who have grown all the three types of crops, and to get to this farmer, they have decided to play a game.

Task

Use the above information to determine if all families in the village will get food.

Part II

Answer only **one item** *from this part.*

Item 5

An events organizing company needs to make a new type of decoration, they want to cut out an octagon out of the circular cardboard whose diameter is 8cm so you are given the task to come up with the accurately cut out decoration.

The head decorator has got a contract to organize an event but he does not have enough money for materials he needs so he wants to borrow shs 8,400,000, the bank has two options but, option 1, they can give him the loan at a compound interest rate of 8% per annum to be paid in 2 years while option 2, the loan can be given at a simple interest rate of 10.5% per month, the bank requires that for both options the money should be paid back in 20 equal installments. He plans to pay back the money in 2 years so he needs your help to decide which option to use.

Task

(a) You are required to show how you were able to draw the accurate octagon as instructed.
(b) Using your mathematical skills recommend a loan option for the head decorator to use.

Item 6

National Drug Authority manager is to deliver drugs from their store to three hospitals. The car he is going to use has 15 litres of fuel and it uses 1 litre of fuel for every 15km. He left the store to the first hospital which is 60km away in a direction of $N70^0E$. He then moved southwards at an average speed of $100kmhr^{-1}$ for 30 minutes to the second hospital. From there, he took a bearing of 200^0 and moved a distance of 40km to the third hospital.

At the third hospital he was shown a direct route that would take him to the store but he was not sure whether the remaining fuel in the car was enough for that journey since he didn't know how long that route was.

Task:

Help the manager to determine;

(a) The distance of the direct route from the third hospital to the store.
(b) Whether the remaining fuel in the car will be enough for the journey using the direct route.

END